# Jannatul Ferdous Srabonee

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As a PhD candidate in Electrical Engineering and Computer Science at UC Merced, my research focuses on designing and developing inclusive input and interaction techniques for digital systems. My current work spans brain-computer interfaces, applied artificial intelligence, gesture mapping, and eye tracking. I am especially interested in extending this work toward adaptive AR/VR interfaces that adjust to a user's perceptual and physical needs.

## Education

**Ph.D. | Electrical Engineering and Computer Science** **2023 – Present**
*University of California, Merced*
Advisor: Dr. Ahmed Arif
Dissertation: Novel Input and Interaction Techniques for Motor and Vision Impaired Individuals

**B.Sc. | Computer Science and Engineering** **2016 – 2020**
*Khulna University of Engineering and Technology, Khulna, Bangladesh*
Advisor: Dr. MAH Akhand
Thesis: EEG based Alcoholism Detection Using CNN through Correlation- based transformation of Channel's Data

## Research Experience

**Graduate Student Researcher** **2023 – Present**
Inclusive Interaction Lab, UC Merced

**Research Engineer** **2021 – 2023**
Advanced Intelligent Multidisciplinary Systems Laboratory, Dhaka, Bangladesh

**Undergraduate Researcher** **2019 – 2020**
Intelligence Systems Lab KUET, Khulna, Bangladesh

## Publications & Manuscripts

### Conference Papers

[C.3] BezelBraille: A Braille-Inspired Bezel-Guided Virtual Keyboard for Low Vision and Blind Users. Isna Alfi Bustoni, **Jannatul Ferdous Srabonee**, Gulnar Rakhmetulla, Ahmed Arif. MobileHCI 2026 [Accepted, not published yet]

[C.2] Exploring the Impacts of HEXACO Personality Traits on Text Composition and Transcription. **Jannatul Ferdous Srabonee**, Ohoud Mosa Alharbi, Wolfgang Stuerzlinger, and Ahmed Arif. CHI 2025 [Paper][Presentation]

[C.1] EEG Based Autism Detection Using CNN Through Correlation Based Transformation of Channels' Data. Zahrul Jannat Peya, MAH Akhand, **Jannatul Ferdous Srabonee**, Nazmul Siddique. IEEE TENSYMP 2020 [Paper]

### Journal Articles

[J.2] Computer Vision-Based IoT Architecture for Post COVID-19 Preventive Measures. Ahsanul Akib, Kamruddin Nur, Suman Saha, **Jannatul Ferdous Srabonee**, M. F. Mridha.

Journal of Advances in Information Technology 2023 [Article]

[J.1] Autism Detection from 2D Transformed EEG Signal using Convolutional Neural Network. Zahrul Jannat Peya, MAH Akhand, **Jannatul Ferdous Srabonee**, Nazmul Siddique.
Journal of Computer Science 2022 [Article]

**Book Chapter**

[B.1] Alcoholism Detection from 2D Transformed EEG Signal. **Jannatul Ferdous Srabonee**, Zahrul Jannat Peya, MAH Akhand, Nazmul Siddique
Proceedings of International Joint Conference on Advances in Computational Intelligence: IJCACI 2020. Springer. [Chapter]

**Under Review**

[2] Gaze & Gape: Hands-Free Text Entry Using Gaze and Mouth-Open Gestures

[1] GazeDriver: Semi and Fully Autonomous Telepresence Robot Control Using Eye Gaze in Virtual Reality

## Posters & Presentations

[Po.2] "AI in Accessibility" | Research Week | 2026 | University of California, Merced

[Po.1] "EEG-based Major Depressive Disorder (MDD) Identification through Functional Connectivity Analysis" | EBRAINS Workshop: Brain Activity across Scales and Species: analysis of experiments and Simulations, | 2022 | Rome, Italy [Virtual]

[Pr.7] "Exploring the Impacts of HEXACO Personality Traits on Text Composition and Transcription" | CHI 2025 | Yokohama, Japan

[Pr.6] "A Machine Learning Based Depression Screening Tool from EEG Data to Aid Clinicians" | Bangladesh Association of Software Information Services (BASIS) National ICT Awards | 2022 | Dhaka, Bangladesh

[Pr.5] "Machine Learning Based Depression Screening Tool from EEG Data to Aid Clinicians" | International Conference on 4IR (4th Industrial Revolution) for the Emerging Future | 2022 | Dhaka, Bangladesh

[Pr.4] "COVID-19 Defense System" | IEEE YESIST12 Maker Fair | 2021 | Dhaka, Bangladesh

[Pr.3] "Mobile App for Post COVID-19 Measures" | NASA Space Apps Challenge | 2021 | Dhaka, Bangladesh

[Pr.2] "Alcoholism Detection from 2D Transformed EEG Signal" | International Joint Conference on Advances in Computational Intelligence | 2020 | Daffodil International University, Dhaka, Bangladesh

[Pr.1] "EEG Based Autism Detection Using CNN Through Correlation Based Transformation of Channels' Data" | IEEE TENSYMP2020 | Dhaka, Bangladesh

## Awards & Scholarship

[A.2] **EECS PBGA Summer Fellowship** (2024, 2025, 2026), UC Merced

[A.1] **Winner Award** (2022), Bangladesh Association of Software Information Services (BASIS) National ICT Awards, Dhaka, Bangladesh

[S.1] General Scholarship in Higher Secondary School Certificate (H.S.C.) examination (2015), Board of Intermediate and Secondary Education, Dhaka, Bangladesh

## Core Skills

**Research Methods:** Human-subject experimental design, User studies & usability testing, Quantitative & qualitative data analysis, Statistical analysis
**HCI & Interaction Design:** Input technique design (gesture, gaze, force-based), Accessible/assistive interface design, Prototyping (low- and high-fidelity)
**Brain-Computer Interfaces & Signal Processing:** EEG signal acquisition & preprocessing, Motor imagery classification, Signal-to-image transformation techniques
**Machine Learning & AI:** TensorFlow, scikit-learn, Computer vision (object/gesture recognition)
**Programming:** Python, MATLAB, C++, Git/version control

## Professional Affiliations

**ACM ASSETS** **Since March, 2026**
Program Committee Member
**Health Sciences and Research Institute at UC Merced** **Since November, 2024**
Student Member
**Association for Computing Machinery (ACM)** **Since September, 2023**
Student Member

## References

**Dr. Ahmed Arif**
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